Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 70 (2019)	31-Jan-2019

S(n)=7400 *SY*; S(p)=2760 *SY*; Q(α)=11000 *70* 2017Wa10
Estimated uncertainties (2017Wa10): ΔS(n)=820, ΔS(p)=1040.
S(2n)=13980 (theory,1997Mo25). S(2p)=4420 *1040*, Q(εp)=340 *880* (syst,2017Wa10).
2012Og06, 2006Og05, 2004Og10 (also 2011Og07, 2009Og07, 2007Og05, 2005Og03, 2005OgZZ, 2004OgZZ, 2004Mo15, 2003OgZZ, 2002OgZX): ^{290}Lv produced as α daughter of ^{294}Og in four correlated decay chains. See ^{294}Og Adopted Levels for details.

2006Og05,2004Og07 (also 2004Og05): ^{290}Lv produced in ^{245}Cm(^{48}Ca,3n),E=255, 249, 243 MeV reaction at FLNR-JINR-Dubna facility using DGFRS separator, optimized for the production of ^{290,291}Lv. The typical beam intensity was 1.2 pμA. The target was enriched to 98.7%. Collaboration of JINR, Russia, and LLNL, USA. A total of 12 correlated decay chains was observed: nine by 2006Og05 and three in 2004Og07 with the following history:

Event #1: Energy of the evaporation residue (EVR)=11.7 MeV (2006Og05).
$E_{\alpha 1}$=11150 keV *310*, $\Delta t_{\alpha 1}$=0.207 ms, assigned to ^{290}Lv.
$E_{\alpha 2}$=10080 keV *310*, $\Delta t_{\alpha 2}$=107.8 ms, assigned to ^{286}Fl α decay.
E_{SF}=170 MeV, Δt_{SF}=1.929 ms, assigned to ^{282}Cn SF decay.

Event #2: Energy of the evaporation residue=10.2 MeV (2006Og05).
$E_{\alpha 1}$=10810 keV *120*, $\Delta t_{\alpha 1}$=20.1 ms, assigned to ^{290}Lv.
$E_{\alpha 2}$=10050 keV *120*, $\Delta t_{\alpha 2}$=54.1 ms, assigned to ^{286}Fl α decay.
E_{SF}=188 MeV, Δt_{SF}=1.636 ms, assigned to ^{282}Cn SF decay.

Event #3: Energy of the evaporation residue=13.3 MeV (2006Og05).
$E_{\alpha 1}$=10800 keV *500*, $\Delta t_{\alpha 1}$=10.0 ms, assigned to ^{290}Lv.
$E_{\alpha 2}$=10190 keV *110*, $\Delta t_{\alpha 2}$=137.9 ms, assigned to ^{286}Fl α decay.
E_{SF}=192 MeV, Δt_{SF}=1.020 ms, assigned to ^{282}Cn SF decay.

Event #4: Energy of the evaporation residue=12.7 MeV (2006Og05).
$E_{\alpha 1}$=10890 keV *80*, $\Delta t_{\alpha 1}$=5.251 ms, assigned to ^{290}Lv.
$E_{\alpha 2}$=10200 keV *500*, $\Delta t_{\alpha 2}$=40.3 ms, assigned to ^{286}Fl α decay.
E_{SF}=193 MeV, Δt_{SF}=1.929 ms, assigned to ^{282}Cn SF decay.

Event #5: Energy of the evaporation residue=11.3 MeV (2006Og05).
$E_{\alpha 1}$ missing α.
$E_{\alpha 2}$=10220 keV *110*, $\Delta t_{\alpha 2}$=62.9 ms, assigned to ^{286}Fl, α decay.
E_{SF}=209 MeV, Δt_{SF}=0.746 ms, assigned to ^{282}Cn SF decay.

Event #6: Energy of the evaporation residue=11.2 MeV (2006Og05).
$E_{\alpha 1}$ missing α.
E_{SF}=176 MeV, Δt_{SF}=876.5 ms, assigned to ^{286}Fl SF decay.

Event #7: Energy of the evaporation residue=11.7 MeV (2006Og05).
$E_{\alpha 1}$=10870 keV *110*, $\Delta t_{\alpha 1}$=0.792 ms, assigned to ^{290}Lv.
$E_{\alpha 2}$=10230 keV *110*, $\Delta t_{\alpha 2}$=339.3 ms, assigned to ^{286}Fl α decay.
E_{SF} missing SF.

Event #8: Energy of the evaporation residue=8.4 MeV (2006Og05).
$E_{\alpha 1}$ missing α.
E_{SF}=203 MeV, Δt_{SF}=252.3 ms, assigned to ^{286}Fl SF decay.

Adopted Levels (continued)

Event #9: Energy of the evaporation residue=14.7 MeV (2006Og05).
$E_{\alpha 1}$=10600 keV *500*, $\Delta t_{\alpha 1}$=22.29 ms, assigned to ^{290}Lv.
E_{SF}=205 MeV, Δt_{SF}=126.4 ms, assigned to ^{286}Fl SF decay.

Event #10: Energy of the evaporation residue=12.7 MeV (2004Og07).
$E_{\alpha 1}$=10880 keV *80*, $\Delta t_{\alpha 1}$=0.233 ms, assigned to ^{290}Lv.
E_{SF}=193 MeV, Δt_{SF}=328 ms, assigned to ^{286}Fl SF decay.

Event #11: Energy of the evaporation residue=15.6 MeV (2004Og07).
$E_{\alpha 1}$ missing α.
E_{SF}=176 MeV, Δt_{SF}=914 ms, assigned to ^{286}Fl SF decay.

Event #12: Energy of the evaporation residue=11.9 MeV (2004Og07).
$E_{\alpha 1}$ missing α.
$E_{\alpha 2}$=10030 keV *310*, $\Delta t_{\alpha 2}$=505 ms, assigned to ^{286}Fl α decay.
E_{SF}=169 MeV, Δt_{SF}=1.448 ms, assigned to ^{282}Cn SF decay.

For theoretical studies, consult Nuclear Science References (NSR) database at NNDC, BNL for 142 primary references dealing with the half-lives and other aspects of nuclear structure in this mass region.

^{290}Lv Levels

Cross Reference (XREF) Flags

A ^{294}Og α decay (0.58 ms)

E(level)	J^π	$T_{1/2}$	XREF	Comments
0	0^+	8.3 ms *+35−19*	A	%α=100 (2016Ho09); %SF≤8 (2016Ho09) Only the α decay mode has been observed. E(level): the reported activity is assumed to belong to the g.s. of ^{290}Lv. $T_{1/2}$: 8.3 ms *+35−19* (2017Og01,2015Og05 reviews); 8.3 ms *+36−19* (2016Ho09 review). In 2016Ho09 review, two separate half-lives of 0.32 ms *+26−10* and 14.9 ms *+103−10* are assigned based on the statistical analysis of all the events observed by the Dubna group. Measurements: 8.3 ms *+35−19* (2012Og06, from 11 events, one in 2012Og06 and 10 in 2006Og05). Others: 7.1 ms *+32−17* (2006Og05,2011Og07); 15 ms *+26−6* (2004Og07,2004Og12,2005Og03 from two events); 29 ms *+140−33* (2002OgZX). Eα=10.85 MeV *7* (2017Og01,2015Og05 reviews; 10.84 MeV *8* in 2011Og07 review); 10.85 MeV *6* (2016Ho09 review). Measurements: 10850 *70* (2012Og06, from a total of 11 events); 10840 *80* (2006Og05). Production σ=3.7 pb *+36−18* at E*=37.9 MeV; 1.9 pb *+21−10* at E*=42.7 MeV (2006Og05); ≈1 pb at E*=(2004Og07).